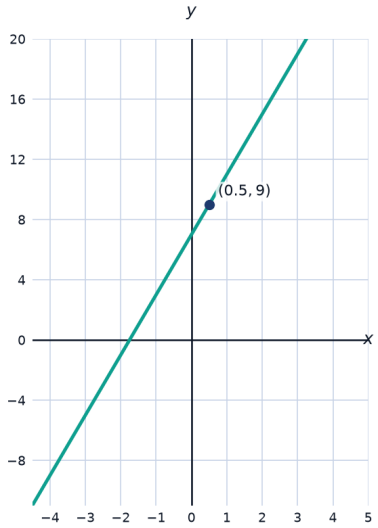
Factoring Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

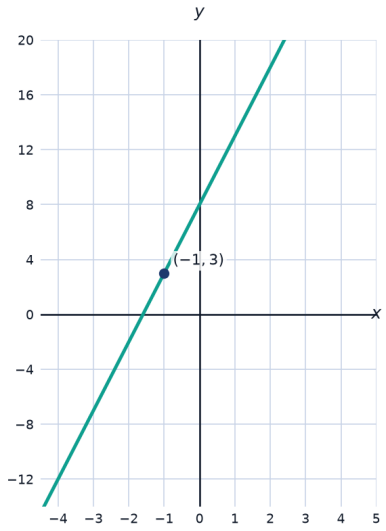
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in solving systems of equations with the elimination method, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 8$ in solving systems of equations with a variety of methods, evaluate $f(2)$.



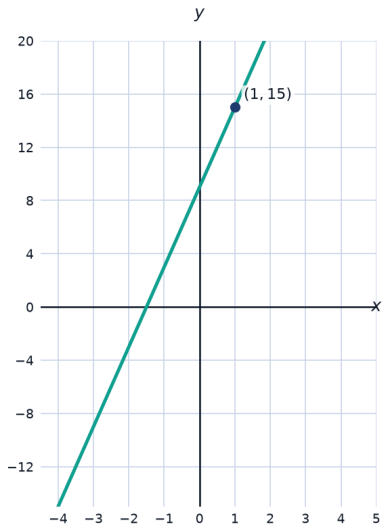
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving systems of equations with a variety of methods consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 9$ in applications of linear systems, evaluate $f(2)$.

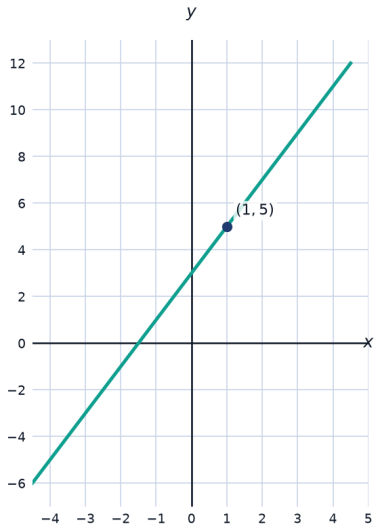


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for applications of linear systems consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 3$ in systems of inequalities, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for systems of inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

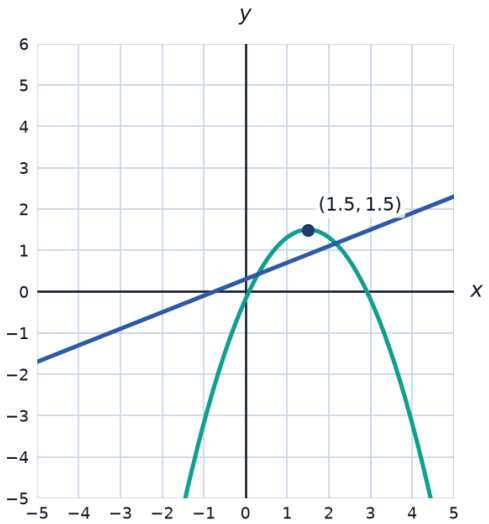
Level 2 · MediumA sample of a radioactive substance decays such that its mass is halved every 15 days. If the initial mass is 480 grams, which function models the mass $M(t)$, in grams, remaining after t days?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 4 · SynthesisA radioactive isotope decays by 20% every 15 years. If the initial mass is 500 grams, which expression represents the remaining mass, in grams, after m months?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 1 · FoundationsWhich expression is equivalent to $\frac{x^{5/3}}{x^{2/3}} \cdot \sqrt{x}$ for all positive values of x?



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A school theater sold 250 tickets for a play, raising a total of $1,750$. $Adulttickets cost$9 each and student tickets cost 5 each. How many adult tickets were sold?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratic Functions

Write the general circle equation $x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, identify its center and radius, and find the equation of the tangent line to the circle at the point $P(4, 1)$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Exponents & Polynomials

Determine the equation of the circle in the xy-plane for which the segment connecting $A(2, -3)$ and $B(8, 5)$ serves as a diameter.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Functions & Transformations

A fair die is rolled twice. Find $P(\text{both show } 6)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Suppose y varies directly as x^2 and inversely as z. If $y = 18$ when $x = 3$ and $z = 4$, find the value of y when $x = 5$ and $z = 10$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Systems & Inequalities

A car travels at 60 mph for 2.5 hours. Find the distance.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

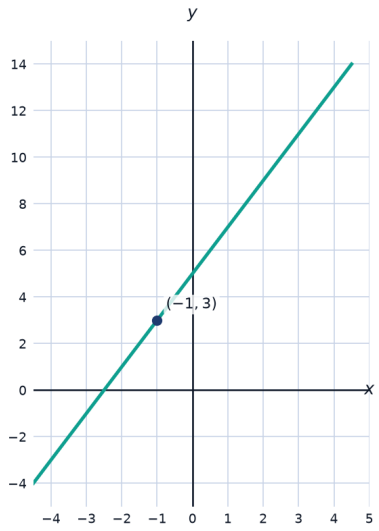
A jacket costs USD 50 after a 20% discount. Find the original price.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in graphing systems of equations, evaluate $f(2)$.



a. 9

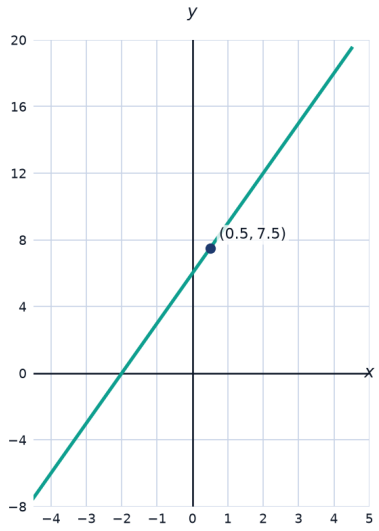
b. 11

c. 4

d. 13

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in solving systems of equations with the substitution method, evaluate $f(2)$.



a. 12

b. 15

c. 6

d. 16

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for graphing systems of equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solving systems of equations with the substitution method consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

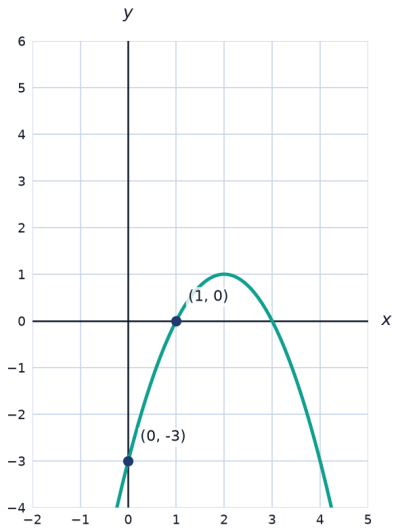
b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.



a. 3

b. 4

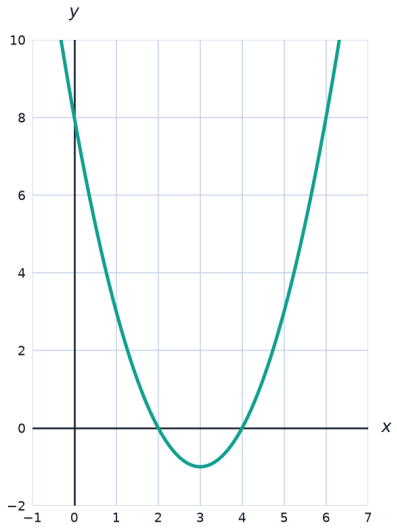
c. 5

d. 6

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the quadratic function $y = x^2 - 6x + 8$ is shown below. What are the coordinates of the vertex of the parabola?



a. (3, -1)

b. (3, 1)

c. (1, 3)

d. (-1, 3)

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the polynomial function $P(x) = x^3 - 6x^2 + 11x - 6$. If $P(x)$ can be factored into the form $(x - a)(x - b)(x - c)$, where $a, b,$ and c are distinct real numbers, what is the value of $a^2 + b^2 + c^2$?

a. 14

b. 29

c. 38

d. 41

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A biologist is modeling the population of a rare plant species. The population, $P(t)$, in thousands, after t years is approximated by the function $P(t) = a \cdot 2^{kt}$, where a and k are constants. If the initial population in year $t = 0$ was 10,000 plants and after 3 years the population had grown to 80,000 plants, what is the approximate population after 6 years, in thousands?

a. 320

b. 640

c. 1280

d. 2560

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

a. -2

b. -0.5

c. 0.5

d. 2

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

A farmer wants to fence a rectangular field adjacent to a straight river. The farmer has 120 meters of fencing material. The side of the field parallel to the river does not require fencing. Let the dimensions of the field perpendicular to the river be x meters, and the dimension parallel to the river be y meters. The area of the field is given by $A(x) = 120x - 2x^2$. What is the maximum possible area of the field?

a. 1800 square meters

b. 2000 square meters

c. 2400 square meters

d. 2800 square meters

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of the function $f(x) = ax^2 - 8x + c$ has its vertex at $(2, -4)$. What is the value of $a + c$?

a. -6

b. -5

c. 5

d. 6

32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The parabola has its vertex at $(-2, 8)$ and passes through the point $(0, 4)$. Determine the value of c.

a. 4

b. -2

c. 8

d. 2

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the function $y = \sqrt{x - a} + b$ is shown below. If the vertex of the graph is at $(7, 3)$, what is the value of $a + b$?

a. 3

b. 4

c. 10

d. 11

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the sum of the roots of the quadratic equation $3x^2 - 11x + 2 = 0$?

a. $-\frac{2}{3}$

b. $-\frac{11}{3}$

c. $\frac{11}{3}$

d. $\frac{2}{3}$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A psychologist conducted an experiment where 80 participants were randomly assigned to either meditate for 15 minutes before an exam or sit quietly. The meditation group scored significantly higher on average. To what population can the causal conclusion be safely generalized?

a. All college students worldwide

b. All adults of any age

c. Individuals similar to the participants in the study

d. No causal conclusions can be made

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $\sqrt{2x + 15} - x = 0$, what is the sum of all valid real solutions?

a. 3

b. 5

c. 7

d. 8

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of x if $\frac{2^x}{2^3} = 16$?

a. 4

b. 5

c. 7

d. 19

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the extraneous solution to the radical equation $\sqrt{x+18} = x-2$?

a. -2

b. -3

c. 5

d. 7

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Systems & Inequalities

For what positive value of c does the equation $2x^2 - 4x + c = 0$ have exactly one distinct real solution?

a. 16

b. 6

c. 2

d. 4

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Sequences & Series

What are all real solutions to the rational equation $\frac{x}{x-3} = \frac{3}{x-3} + 4$?

a. $x = 3$

b. $x = 4$

c. $x = 7$

d. No real solution

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Radicals & Exponents

If $(x - 3)$ is a factor of the polynomial $P(x) = 2x^3 - kx^2 + 1x - 3$, what is the value of the constant k?

a. 6

b. 3

c. 8

d. -6

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Linear Equations & Lines

What is the valid real solution to the equation $\frac{x}{x-2} + \frac{1}{x+2} = \frac{8}{x^2-4}$?

a. -5

b. -2

c. 2

d. 5

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Rational Expressions

If $x = 3$ is a solution to the quadratic equation $2x^2 + kx - 15 = 0$, what is the value of the constant k?

a. -1

b. 1

c. -3

d. 3

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Statistics & Data Analysis

Consider the quadratic function $f(x) = x^2 - 6x + 9$. What is the value of $f(3)$?

a. 0

b. 3

c. 9

d. 18

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratic Functions

Section 1 (calculator-allowed) has 24 questions completed in 30 minutes; Section 2 has 20 questions completed in 20 minutes. Keeping each section's original per-question pace, how many total minutes would be needed if each section instead had 30 questions?

a. 62.5 minutes

b. 50.0 minutes

c. 57.5 minutes

d. 67.5 minutes

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